

Jack Ma

rzma0628@gmail.com | jackma.me | github.com/RZ3M | linkedin.com/in/rzma | 647-229-0066 | Toronto, ON

Education

Ontario Tech University – Bachelor of Science (Honours) in Computer Science – **3.86 GPA**

2025

Skills

Languages: Python, SQL, C#, TypeScript, JavaScript, Java, Bash

Frameworks & Libraries: .NET, ASP.NET Core, React, Next.js, Node.js, Express, Spring Boot, REST APIs

AI & Data: RAG, vector search, embeddings, retrieval evaluation, cross-encoder reranking, function calling, structured output, Pandas/NumPy

Databases: SQL Server, MariaDB, MongoDB, Elasticsearch, Supabase, Qdrant

Cloud & Tools: Azure, AWS, Docker, GitHub Actions, Jenkins, Git, Linux, CI/CD, SonarQube

Experience

Dayforce – *Software Developer Co-op (DevOps)* – North York, ON

Sep 2025 – Present

- Cut file load latency from **1 min to under 50 ms** by migrating file management to Azure Blob Storage behind an in-memory blob/file-server resolver cache
- Built a C#/.NET deletion-queue workflow on SQL Server that reclaimed **60+ TB** by retiring **5,000+** orphaned databases
 - Designed the lifecycle (notify, approve, backup, drop, unregister) with per-DB failure isolation
 - Added reservation/login reconciliation and a React approval UI with audit and notification guardrails
- Migrated a core platform integration from a deprecated internal tool to its replacement, refactoring **20+** call sites behind typed DTOs and mockable service abstractions
- Cut internal admin dashboard load times by **99%+ (5-10 min to under 5 sec)** by collapsing N+1 per-service health polls into a single control-database query
- Cut CI pipeline runtime from **30 to 18 min (40% faster)** by mocking live Azure calls, deduping build steps, and parallelizing tests
- **Tech Stack:** C#/.NET, SQL Server, Azure, React, TypeScript, Docker, Moq, SonarQube

Royal Bank of Canada – *Software Developer Co-op (Full-Stack)* – Toronto, ON

Sep 2022 – Apr 2023

- Shipped a React + Redux dashboard for the Enterprise Architecture team indexing **1,000+** approved technologies and services used by teams across the bank, backed by Spring Boot APIs
- Designed Spring Boot REST APIs and tuned MariaDB/Elasticsearch indices, cutting service-metadata reports from **10s to 3s** and reducing manual reporting effort by **80%**
- Upgraded **10+** backend services to current Spring LTS, raising JUnit coverage to **80%**
- **Tech Stack:** Java, Spring Boot, MariaDB, Elasticsearch, React, Redux, TypeScript, JUnit, Mockito, Jenkins, Git

IBM – *Cognos Support Engineer Co-op* – Ottawa, ON

Sep 2019 – Dec 2019

- Debugged **100+** production BI deployment issues by tracing logs, profiling Cognos report queries, and inspecting config/auth layers, maintaining **98%+** uptime across enterprise client environments
- Built internal Bash and PowerShell diagnostic scripts that automated recurring health checks, cutting average triage time from **45 to 15 min** per case
- **Tech Stack:** IBM Cognos, Bash, PowerShell, SQL

Projects

Terraria RAG System

github.com/RZ3M/terraria-rag

- Built a local-first **Python** RAG indexing **3,671** Terraria wiki pages into **22,005** Qdrant chunks for spoiler-free hints
- Hit **81.8% recall@5**, **95.5% content-recall@5**, **0.629 MRR** via metadata filters, section reranking, and cross-encoders

Flash.AI – HackHive 2025 Winner

devpost.com/software/flash-ai-ulfqtp

- Won **HackHive 2025** by building a React/Vite + Express/MongoDB AI study assistant with a Google Gemini pipeline (PDF/TXT/DOCX to tagged, difficulty-rated quizzes)
- Returned to **HackHive 2026** as a mentor representing Dayforce, hosting workshops for student builders